

XRPL Validator — Daily Engineering Report • 2026-06-30

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- 2 commits (xrpl-validator 2)

Engineering changes

- **fix(ff): make CallbackReadView succ upper bound exclusive (open interval)** (6893bbe , xrpl-validator)
rippled's ReadView::succ contract is the OPEN interval (key, last). The OverlayedDbProvider::succ callback treated `last` as inclusive ($k \leq \text{last}$) in both the snapshot scan and the overlay filter. Dormant for owner-dir / order-book walks (their `last` sentinel never equals a live key) but load-bearing for NFTTokenMint: rippled locates a mint's target page via `succ(first, nftpageMax(owner).next())`, so an inclusive ...
- **fix(ff): make CallbackReadView succ upper bound exclusive (open interval)** (59bcaec , xrpl-validator)
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XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: [iceland.halcyon-names.io](#)). They are exploration leads, not proposals.

- **Continuously-vesting token payment streams** (payments) Add a `Stream` ledger object (`ltSTREAM` , owner-reserved, `keylet::stream(account, sequence)`) and three transactors. `StreamCreate` locks XRP, an IOU
- **Confidential MPT balances via Groth16 proofs** (privacy) Add an `lsfMPTConfidential` flag to `MPTokenIssuance` and a new `MPTConfidentialSend` transaction; in confidential mode each `MPToken` stores ``sfBalan`
- **Account-level outbound spend policies** (accounts) Add an `AccountPolicy` ledger object (one per account, owner-reserved, `keylet::accountPolicy(account)`) set via a new `AccountPolicySet` transaction
- **Rentable Time-Limited NFTs** (nft) Reuse the NFTTokenOffer machinery: add a `tfLease` flag to `NFTTokenCreateOffer` and a new inner `SField sfExpiration` (UInt32 close-time) on the offer template
- **Credential-Gated Token Transfers** (compliance) Add an optional `sfDomainID` (Hash256 keylet of a `PermissionedDomain`) to `MPTokenIssuance`. When set, the MPT transfer path (`Payment / MPTSend`) gains a pr

- **Native Oracle-Triggered Stop Orders** (dex) Add a tfTrigger flag to OfferCreate plus three SFields — sfTriggerOracle (Hash256 PriceOracle keylet), sfTriggerThreshold (Number), and sfTriggerDir (
- **Time-Locked Social Guardian Account Recovery** (accounts) Add a RecoverList ledger object (ltRECOVER_LIST, owner-reserved, keylet::recover(account)) set by a new RecoverSet transaction that registers guardian
- **Fee And Reserve Sponsorship For Onboarding** (fees) Add two optional common transaction fields, sfSponsor (AccountID) and sfSponsorSignature; when present, preclaim verifies the sponsor signed over the
- **Consensus-Native Randomness Beacon** (oracles) Add a RandomCommit ledger object (ltRANDOM, owner-reserved, keylet::random(account, sequence)) created by a RandomCommit transaction recording sfNonce
- **Atomic post-recovery key hygiene** (accounts) On RecoverFinish doApply, walk the owner directory to delete the account's SessionKey (g11) SLEs and reset any AccountPolicy (g18) spend-limit state i
- **Royalty-Locked NFTs: Ban Zero-Value Sell Offers** (nft) Add a new NFTTokenMint flag tfTransferFeeLocked (packed as lsfTransferFeeLocked in the existing NFTTokenID flag nibble, valid only when sfTransferFee >
- **Bounded Batch Sweep for Dust Objects** (fees) Add a new transaction ttOBJECT_SWEEP carrying sfSweepIDs, a bounded STVector256 (max 16 entries) of ledger-object keylets owned by the submitting Acco
- **Deterministic Volatility Surcharge for AMM Fees** (amm-defi) Add two SFields to the AMM ledger object: sfPriorClosingPrice, a spot-price snapshot (LP-token-implied ratio) taken automatically at the first swap to
- **Directory-node-touch metered fee primitive** (aidea) Generalize the critical analysis's mispricing finding into a reusable fee primitive: meter the actual number of owner-directory page inserts/merges/de
- **Pluggable deletability-predicate registry** (aidea) Extract g27's six inline 'safe to remove' checks into a small on-ledger-agnostic registry keyed by object type, so any future amendment (starting with

Appendix — commits

Time	Repo	Commit	Subject
17:51	xrpl-validator	6893bbe	fix(ffi): make CallbackReadView succ upper bound exclusive (open interval)
18:08	xrpl-validator	59bcaec	fix(ffi): make CallbackReadView succ upper bound exclusive (open interval)